Plan the delivery route

Name: ______________________________

Date: ____________________ **Period:** ____________________

Today I can...

Put steps in order, then test and adjust a short route.

Words to know

Sequence: Steps in order.

Calibrate: Test and adjust the settings.

Variable: One thing you can change, such as drive time.

LOOK AT THE IDEA

1. FORWARD → **2. RIGHT** → **3. FORWARD** → **4. STOP**

Test each part. Stop between parts. Then join the steps.

Gather your materials

- Checked base, ruler and removable floor tape
- Six paper cards about 8 cm square
- Pencil or marker
- The short-drive code from lesson 2

Build a route with arrow cards

Build with the power off. Tick each step when it is done.

1. Draw forward arrows on four cards, a right-turn arrow on one card and STOP on the last card. These cards are a plan; they do not control the robot by themselves.

2. Mark a short straight lane and a wide delivery space on the floor. Leave enough clear room for a turn. Keep arrow cards beside the lane, away from the wheels.

3. Arrange the cards: drive forward, turn right, drive forward, stop. Point to the direction the robot will face after each step.

4. Test the first short drive on its own. Adjust its time until it reaches the turning area. Calibrate means test and adjust a setting.

Sketch the setup. Label the parts you will check.

Finish the setup

5. With the teacher, test a short, slow right turn by itself. Stop after the turn. Change its time a little if it turns too far or not far enough.
6. Test the second short drive. Join the three checked parts, with a Stop after each one. Keep the first complete tests to 3 seconds of movement or less.
7. Run the route three times from the same mark and direction. Record where the robot stops each time. Keep people and objects out of the lane.
8. Change one time setting and try another set of runs. Save the code and draw the final route.

☐ Teacher check: parts secure, wheels and sensor clear, Stop ready, test area clear. Power off before changing parts.

Show where your robot starts and how you will stop it.

Code it. Predict. Try it.

Make the program

1. Start stopped. Drive slowly for the tested first time, then stop.

2. Turn right slowly for the tested turn time, then stop.

3. Drive slowly for the tested second time, then stop and end.

Coding Canvas is the app where you make the program. Use your teacher-checked settings before a floor run.

Before you run: what do you think will happen?

After one try: what actually happened?

End of class 1: save the code, stop and power off, label your parts, and keep these pages.

Test it fairly

Name: ______________________________

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Change one time setting. Keep the speed, floor, load and starting direction the same.

A successful test

- ☐ The program stops after each part and at the end.
- ☐ The robot finishes inside the space in at least two of three runs.
- ☐ The team records the times and layout so it can try again.

Record every result

Run	Drive / turn / drive (seconds)	Inside space?	What happened?
1			
2			
3			

Which result stands out?

____

Change one thing. Try again.

The one thing we changed:

We kept these things the same:

Run	Drive / turn / drive (seconds)	Inside space?	What happened?
1			
2			
3			

Which time setting did you change, and what happened?

Use a result: our change helped / did not help because...

Before leaving: save your code, stop and power off the robot, return parts, and hand in this module.